

Sheet 11

1.1 $E_{LS}(\vec{w}) = \frac{1}{2} \sum_{i=1}^n (\vec{w}^T \vec{x}_i - \vec{y}_i)^2 \stackrel{!}{=} \frac{1}{2} (\mathbf{X} \vec{w} - \mathbf{y})^T (\mathbf{X} \vec{w} - \mathbf{y})$

$$\begin{aligned} (\mathbf{X} \vec{w} - \mathbf{y})_i &= X_{ij} w_j - y_i \\ &= (x_i)_j w_j - y_i \\ &= \vec{w}^T \vec{x}_i - \vec{y}_i \end{aligned}$$

1.2

sym. Matrix: $A = (a_{ij})$

$$(\vec{w}^T A \vec{w}) = w_i A_{ij} w_j$$

$$\begin{aligned} \frac{\partial}{\partial w_k} w_i A_{ij} w_j &= A_{ij} w_j \delta_{ki} + w_i A_{ij} \delta_{jk} \\ &= A_{kj} w_j + w_i A_{ik} \end{aligned}$$

$$\rightarrow \nabla_{\vec{w}} (\vec{w}^T A \vec{w}) = \sum_{k=1}^n A_{kj} w_j + w_i A_{ik} = A \vec{w} + A^T \vec{w}$$

$$\begin{aligned} \rightarrow \text{if } A \text{ sym.} \hookrightarrow A_{ij} &= 0 \text{ if } i \neq j \Rightarrow \nabla_{\vec{w}} (\vec{w}^T A \vec{w}) = \sum_{k=1}^n A_{kk} w_k + w_k A_{kk} \\ &= 2 A \vec{w} \end{aligned}$$

1.3 Minimized if $\nabla_{\vec{w}} E_{LS}(\vec{w}^*) \stackrel{!}{=} 0$

$$\begin{aligned} E_{LS}(\vec{w}) &= \frac{1}{2} (\mathbf{X} \vec{w} - \vec{y})^T (\mathbf{X} \vec{w} - \vec{y}) = \frac{1}{2} (\vec{w}^T \mathbf{X}^T - \vec{y}^T) (\mathbf{X} \vec{w} - \vec{y}) \\ &= \frac{1}{2} (\underbrace{\vec{w}^T \mathbf{X}^T \mathbf{X} \vec{w}}_{\alpha} - \underbrace{\vec{y}^T \mathbf{X} \vec{w}}_{\alpha} - \underbrace{\vec{w}^T \mathbf{X}^T \vec{y}}_{\alpha^T} + \vec{y}^T \vec{y}) \end{aligned}$$

since α is a scalar: $\alpha = \alpha^T$

$$E_{LS}(\vec{w}) = \frac{1}{2} (\vec{w}^T \mathbf{X}^T \mathbf{X} \vec{w} - 2\alpha + \vec{y}^T \vec{y})$$

$$\begin{aligned} \nabla_{\vec{w}} E_{LS}(\vec{w}) &= \frac{1}{2} (2 \underbrace{\mathbf{X}^T \mathbf{X}}_{\text{sym.}} \vec{w} - 2 \mathbf{X}^T \vec{y}) \\ &= (\mathbf{X}^T \mathbf{X} \vec{w} - \mathbf{X}^T \vec{y}) \end{aligned}$$

$$\Rightarrow \nabla_{\vec{w}} E_{LS}(\vec{w}^*) = \underbrace{\mathbf{X}^T \mathbf{X}}_{\mathbf{1}\mathbf{1}} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \vec{y} - \mathbf{X}^T \vec{y} = 0$$